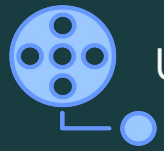
Isolation Forest

F. T. Liu, K. M. Ting, and Z. -H. Zhou (2008)

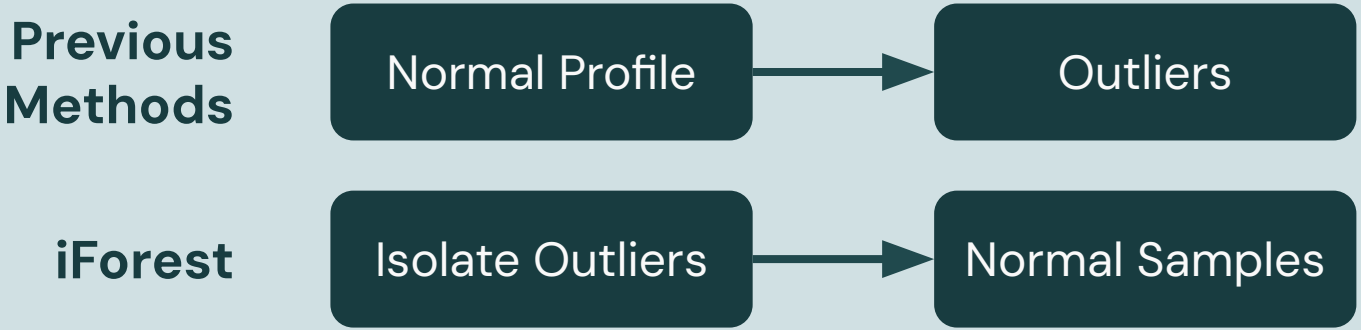
By: Dai Tan Ngo, Jeremy Perkins, Naomi Phan



iForest Overview

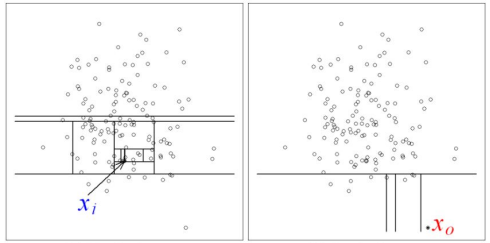


Unsupervised ML Model for Detecting Outliers using Random Trees



Driven by 2 Characteristics: Outliers are Few and Different

iForest Algorithm



(a) Isolating x_i (b) Isolating x_o

Randomly Partition Data

Sub-sampling ($\psi = 256$)

Attribute

Split Value

Construct the Trees

Limit Height = ceiling($\log_2(\psi)$)

Binary Search Tree

Find Average Path Length

$s \in \{0, 1\}$

Anomaly: $s \geq 0.6$

Compute Anomaly Scores

Normalized Avg Path Length

Advantages

Linear Time Complexity

01

Low Constant & Memory Requirement

02

Large & High Dimensional Datasets

03

Training does not Need to Contain Anomalies

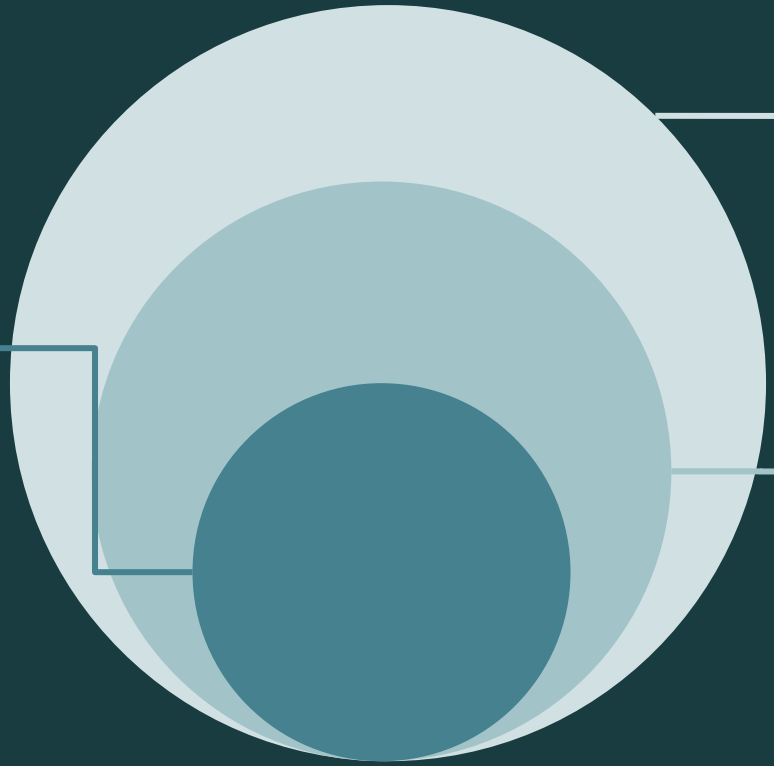
04

eBird Observation Dataset

Training Dataset:
6 Features
1M Samples

1. Taxon Concept ID
2. Species Frequency
3. Observation Count
4. Latitude
5. Longitude
6. Day of year (Sin)
7. Day of year (Cos)

1 = Anomaly; 0 = Not



Original Dataset:
~ 1.5M Samples
~ 10GB
51 Features

Filtered Dataset (PySpark)
12 Features
2GB

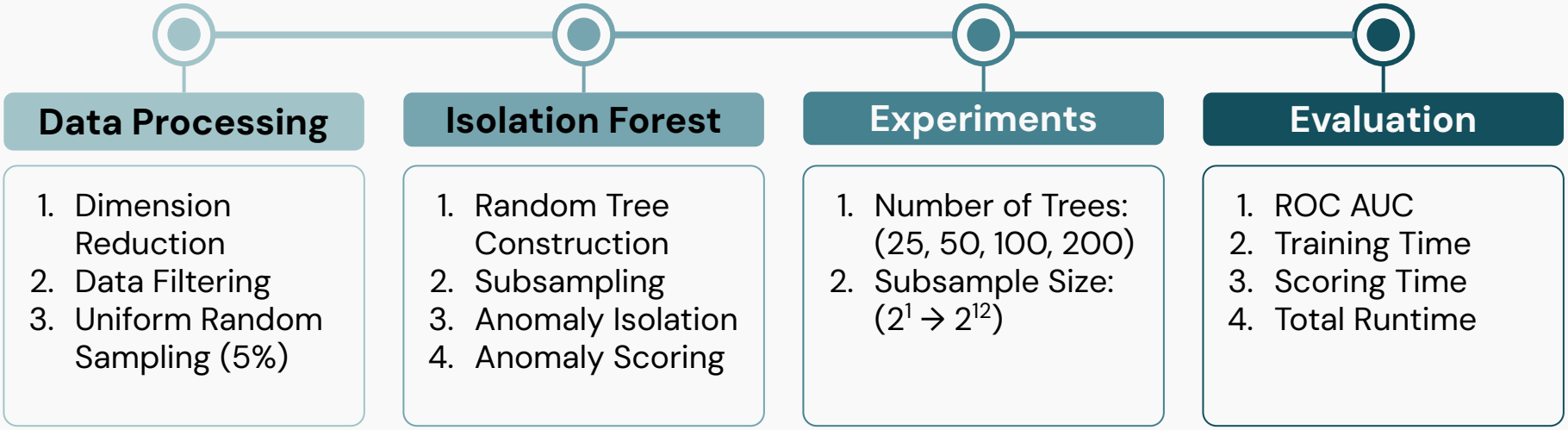
Anomalies in Bird Observation Data

- Geospatial Outlier (Location)
- Temporal Outlier (Time)
- Statistical Outlier (Count)

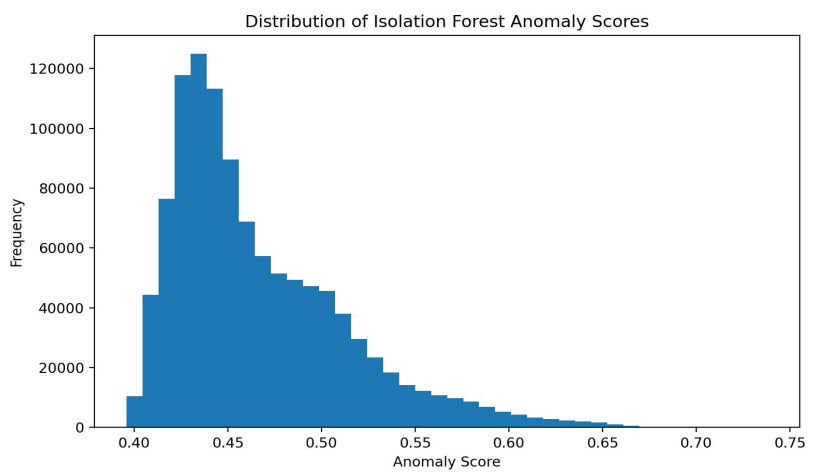
Reviewed and Verified by Experts

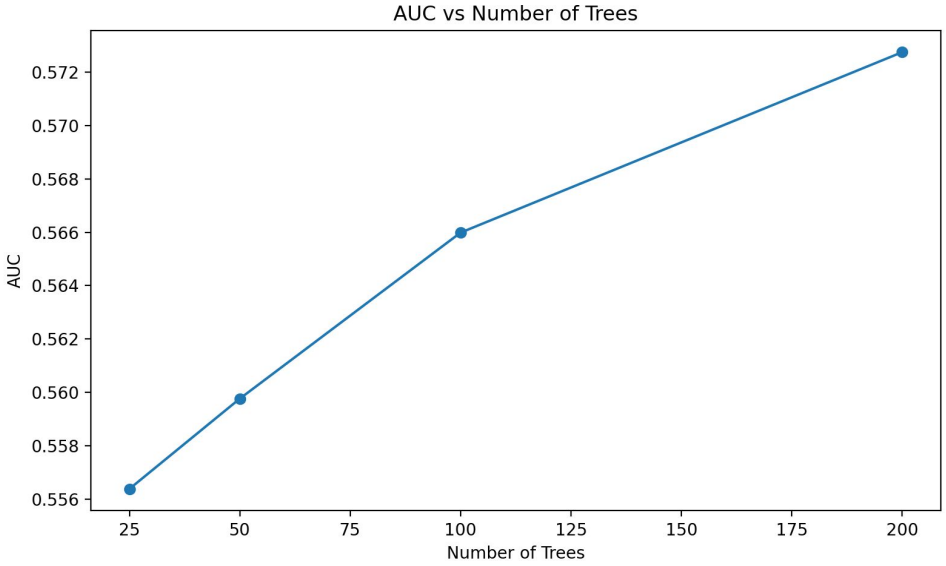


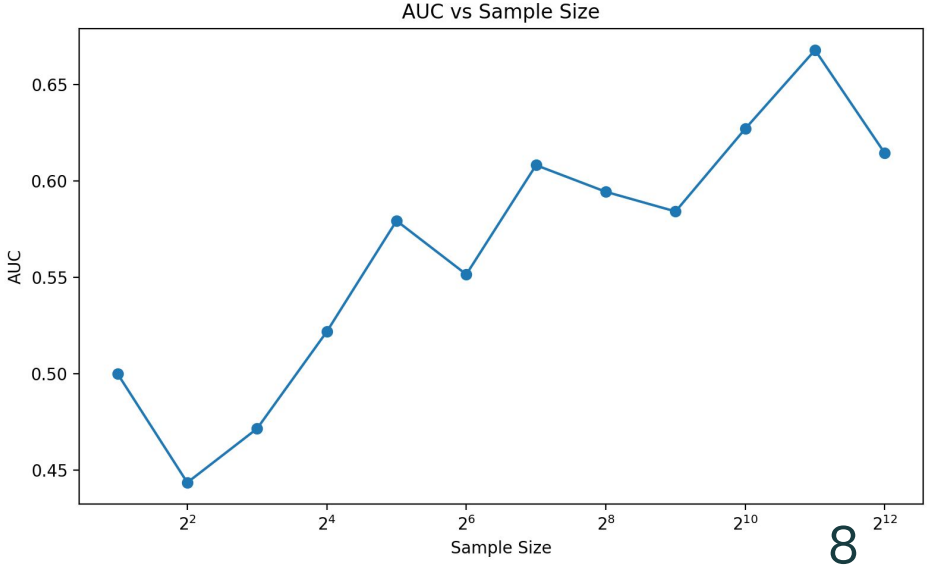
Implementation Pipeline



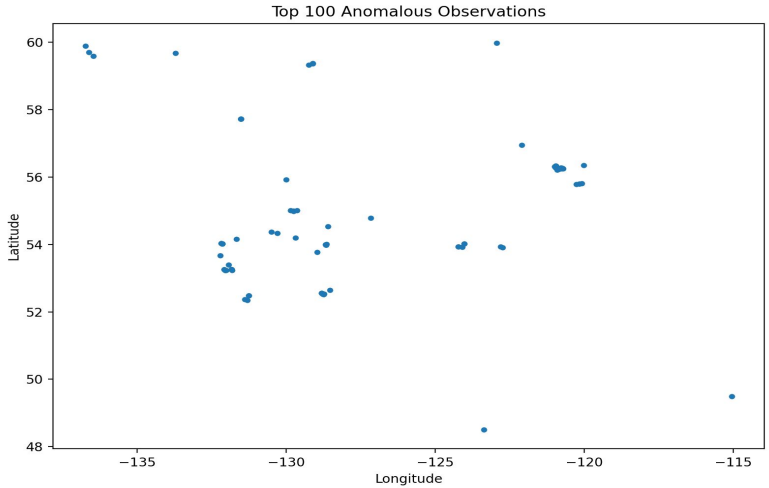
Accuracy Results

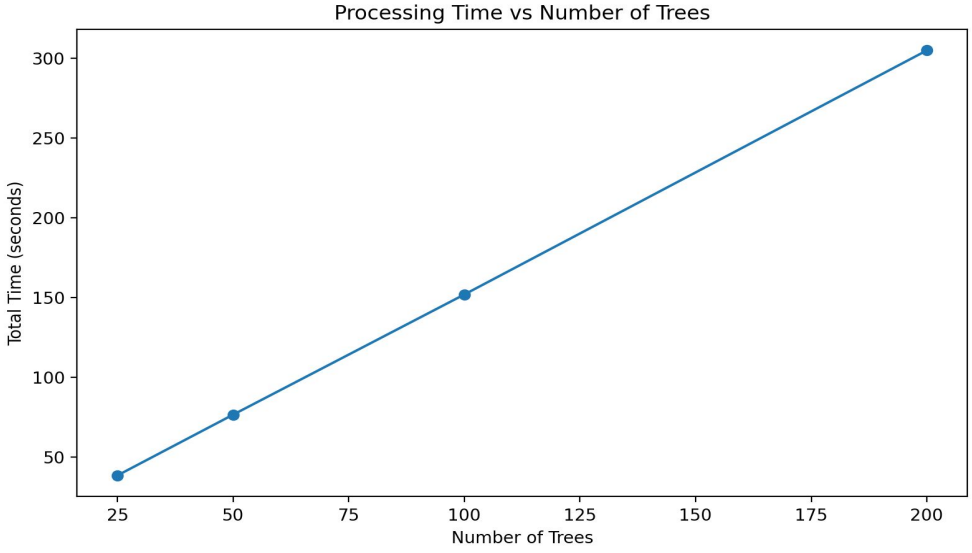


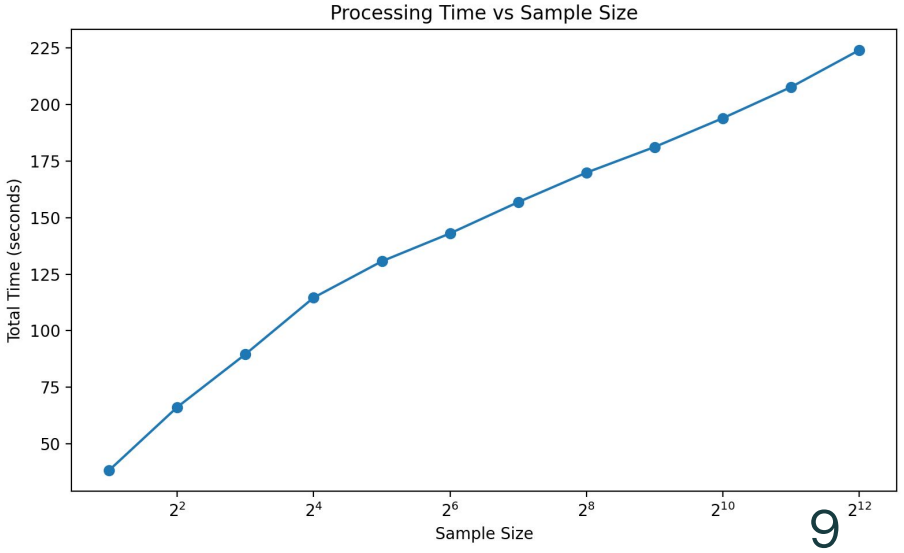




Runtime Results







Future Improvement

01

More Computational Power

02

Incorporate Categorical Features

03

Supervised Learning Injection

04

Diversified Dataset

Thank you for your attention!

Q & A